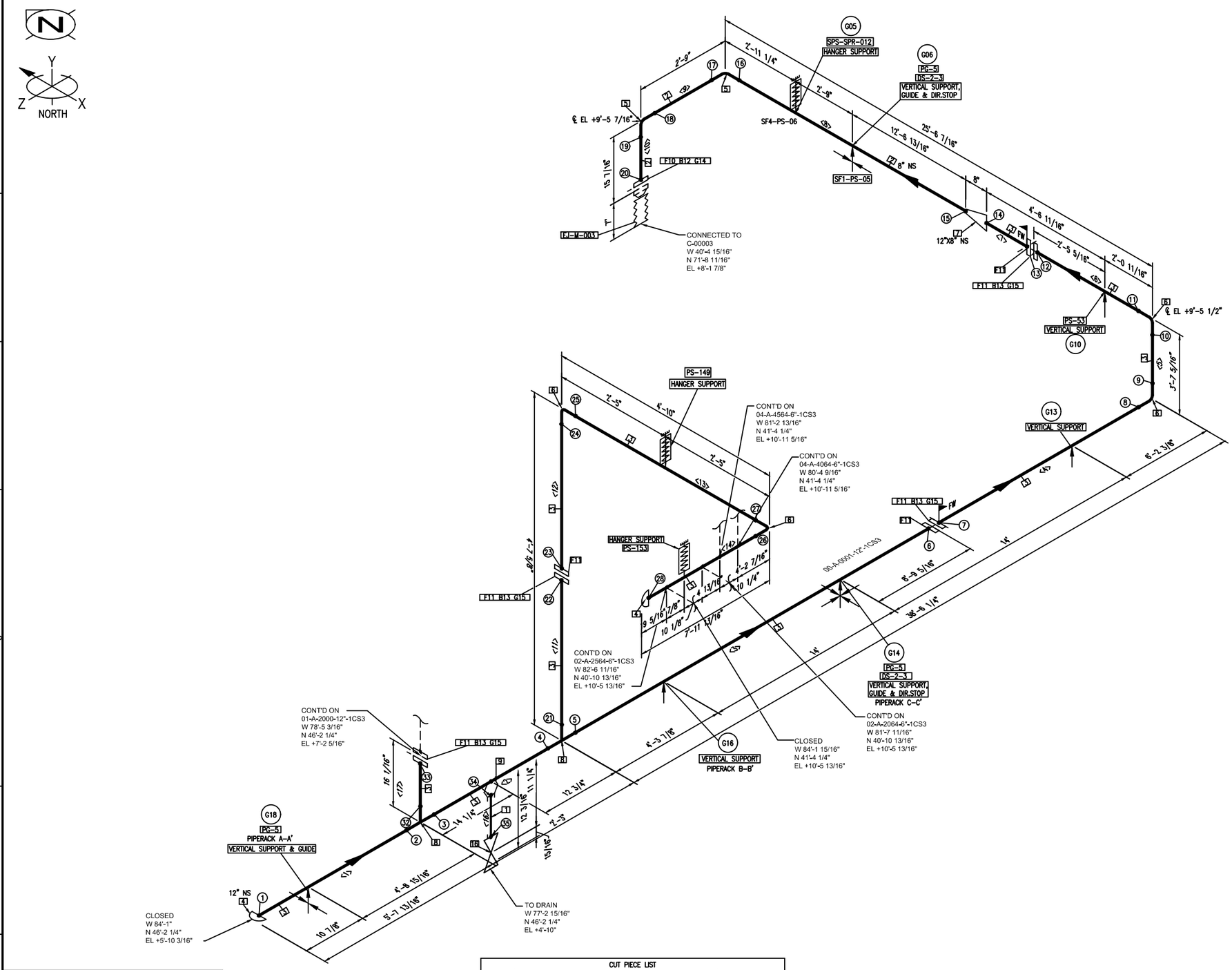


BILL OF MATERIALS			
ID	ND	DESCRIPTION	QTY
1	3/4"	PIPE, XS, CS, SMLS, ASTM A53-B, TYPE S	4"
2	8"	PIPE, STD, CS, ERW, ASTM A53-B, TYPE E (EJ=0.85)	18'-4"
3	12"	PIPE, STD, CS, ERW, ASTM A53-B, TYPE E (EJ=0.85)	58'-4"
4	12"	CAP, WELD, CS, ASTM A234-WPB-S, ASME B16.9	2
5	8"	90 LR ELL, WELD, CS, ASTM A234-WPB-W, ASME B16.9	2
6	12"	90 LR ELL, WELD, CS, ASTM A234-WPB-W, ASME B16.9	4
7	12"X8"	REDUCER (CONC), WELD, CS, ASTM A234-WPB-W, ASME B16.9	1
8	12"	TEE, WELD, CS, ASTM A234-WPB-W, ASME B16.9	2
9	12"X3/4"	SOCKOLET, CLASS 3000, WELD, CS, ASTM A105N, MSS SP-97	1
10	8"	FLANGE SLIP-ON, CLASS 150, RF, CS, ASTM A105N, ASME B16.5	1
11	12"	FLANGE SLIP-ON, CLASS 150, RF, CS, ASTM A105N, ASME B16.5	7
12	3/4"X4 1/4"	STUD BOLT, ASTM A193, GR B7 STUD, W/ 2 HEAVY HEX NUTS, ASTM A194, GR 2H	8
13	7/8"X4 3/4"	STUD BOLT, ASTM A193, GR B7 STUD, W/ 2 HEAVY HEX NUTS, ASTM A194, GR 2H	48
14	8"	GASKET, CLASS 150, 3/16" THK, SPIRAL WOUND TYPE, 304 SS WINDING, W/ FLEXIBLE GRAPHITE FILLER, W/ 304 SS INNER RING AND CS OUTER RING, ASME B16.20	1
15	12"	GASKET, CLASS 150, 3/16" THK, SPIRAL WOUND TYPE, 304 SS WINDING, W/ FLEXIBLE GRAPHITE FILLER, W/ 304 SS INNER RING AND CS OUTER RING, ASME B16.20	4
16	3/4"	GATE VALVE, SW, CLASS 800, CS BODY, TRIM 8	1

MECHANICAL NOTES:-

1. C-0003 INLET NOZZLE LOADS (AT NODE G00) ARE ACCEPTABLE.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES UNLESS NOTE OTHERWISE.
2. FABRICATOR HAS TO VERIFY/ENSURE THE ACTUAL DIMENSIONS BETWEEN INSTALLED EQUIPMENTS & RECEIVED VALVES, INSTRUMENTS & OTHER BOUGHT OUT ITEMS BEFORE FABRICATION OF THIS DRAWING.
3. IN CASE OF PRE FABRICATION SUITABLE SITE ADJUSTMENT PROVISION SHALL BE MADE.
4. REFERENCE PIPING LAYOUT DRAWING NO. :-CED-PIPING LAY-0001.
5. □ DENOTE COMPONENT NUMBER
   <> DENOTE CUTTING PIPE LENGTH
   ⊂⊃ DENOTE TAG NUMBER
   ○ DENOTE WELD/JOINT NUMBER
6. FOR PIPING SUPPORT FABRICATION PLEASE REFER PIPING SUPPORT GA DRAWING AND PIP FABRICATION / INSTALLATION DETAILS.
7. FABRICATION TOLERANCES.
   7.1. LINEAR DIMENSIONS (CENTERLINE TO CENTERLINE).
   - NPS ≤2": ±1/16" (1.5 MM)
   - NPS 3"-12": ±1/8" (3 MM)
   - NPS >12": ±1/4" (6 MM)
   7.2. SPOOL OVERALL LENGTH.
   - UP TO 20FT: ±1/8"
   - OVER 20FT: ±1/4"
   7.3. ANGULAR TOLERANCES.
   - JOINT ANGULAR DEVIATION: ±0.5° PER WELD.
   - ACCUMULATED DEVIATION SHALL NOT IMPAIR FIT-UP OR EXCEED CODE LIMITS.
   7.4. FLANGE ALIGNMENT.
   - FACE TO FACE PARALLELISM: ≤1/16".
   - BOLT HOLE ROTATION: ±1/32" MAX (OR WITHIN BOLT CLEARANCE).
   - FLANGE MISALIGNMENT SHALL COMPLY WITH ASME B31.3
8. PIPE SUPPORT MATERIAL SHALL BE THE SAME AS THE SUPPORTED PIPE.

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NO.	DATE	BY	DESCRIPTION	CKD	APP
E	06/10/2026	MS	ISSUED FOR CONSTRUCTION	RM	MM
D	05/12/2026	SG	ISSUED FOR REVIEW	RM	MM
C	04/08/2026	SG	ISSUED FOR REVIEW	RM	MM
B	01/20/2026	SG	ISSUED FOR REVIEW	RM	MM
A	10/18/2025	SG	ISSUED FOR REVIEW	RM	MM

ISSUE	DATE	Ref.Ind.	Sign/Init.	Ref.Ind.	Sign/Init.	DESCRIPTION
		DESIGNER		DRAFTSMAN		
E	06/10/2026	-	-	MS	-	ISSUED FOR CONSTRUCTION
D	05/12/2026	-	-	SG	-	ISSUED FOR REVIEW
C	04/08/2026	-	-	SG	-	ISSUED FOR REVIEW
B	01/20/2026	-	-	SG	-	ISSUED FOR REVIEW
A	10/18/2025	-	-	SG	-	ISSUED FOR REVIEW
ISSUE	DATE	APPROVED BY				DESCRIPTION

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AVNOS CEDAR HDAC PROJECT

ISOMETRIC DRAWING 00-A-0001-12"-1CS3

LOCATION:	STATION:	PROJECT NO./REQUISITION

REV.:	SCALE:	SHT. NO.:	1 OF 1
E	1:1	DWG. NO.:	00-A-0001-12"-1CS3

WELD LIST		
ID	ND	TYPE
1	12"	BUTTWELD
2	12"	BUTTWELD
3	12"	BUTTWELD
4	12"	BUTTWELD
5	12"	BUTTWELD
6	12"	WELD
7	12"	WELD
8	12"	BUTTWELD
9	12"	BUTTWELD
10	12"	BUTTWELD
11	12"	BUTTWELD
12	12"	WELD
13	12"	WELD
14	12"	BUTTWELD
15	8"	BUTTWELD
16	8"	BUTTWELD
17	8"	BUTTWELD
18	8"	BUTTWELD
19	8"	BUTTWELD
20	8"	WELD
21	12"	BUTTWELD
22	12"	WELD
23	12"	WELD
24	12"	BUTTWELD
25	12"	BUTTWELD
26	12"	BUTTWELD
27	12"	BUTTWELD
28	12"	BUTTWELD
29	12"	TAPWELD
30	3/4"	SOCKETWELD
31	3/4"	SOCKETWELD
32	12"	BUTTWELD
33	12"	WELD
34	3/4"	SOCKETWELD
35	3/4"	SOCKETWELD

CUT PIECE LIST				
ID	LENGTH	ND	END1	END2
1	4'-3 13/16"	12"	BEVEL	BEVEL
2	7"	12"	BEVEL	BEVEL
3	26'-3 1/4"	12"	BEVEL	SO
4	10'-4 5/8"	12"	SO	BEVEL
5	7 5/16"	12"	BEVEL	BEVEL
6	3'	12"	BEVEL	SO
7	2'-6 15/16"	12"	SO	BEVEL
8	17'-3 1/16"	8"	BEVEL	BEVEL
9	9"	8"	BEVEL	BEVEL
10	3 3/16"	8"	SO	BEVEL
11	5 11/16"	12"	BEVEL	SO
12	1'-9 9/16"	12"	BEVEL	SO
13	1'-10"	12"	BEVEL	BEVEL
14	5'-11 7/8"	12"	BEVEL	BEVEL
15	3 5/16"	3/4"	SQUARE CUT	SQUARE CUT
16	3 5/16"	3/4"	SQUARE CUT	SQUARE CUT
17	5 13/16"	12"	SO	BEVEL

PIPE PROPERTIES								
PIPE-ID	NPS	SCH/WALL THK	CORR	INS-THK	SG	MAT	FLANGE RATING	P&ID
0001-CS1S	12	STD	0.063	0.0 IN	1.00	A53-B	150#	PID-0001

PIPING LOADS TABLE													
NODE	THERMAL + GRAV			HYDROTEST	SNOW	ICE	WIND (+/-)			SEISMIC (+/-)			
	Fx [lbs]	Fy [lbs]	Fz [lbs]	FyHT [lbs]	FySN [lbs]	FyICE [lbs]	FxW [lbs]	FyW [lbs]	FzW [lbs]	Fx EQ [lbs]	Fy EQ [lbs]	Fz EQ [lbs]	
G05		-1680		-1210	-90	-80							
G06	-450		1030				-150		-170	-300			-140
G10	-790	-2710	720	-1960	-890	-1420	30	-70	40		-170		
G13	180	-2660	-200	-870		140	-20	-80	-20		-70		
G14	-470	-2370	-660	-2040	-780	-1100	-350	-40	-130	-300	-120		-570
G16	120	-1880	750	-1720	-580	-710	-80				-80		
G18	-120	-800	-320	-790	-220	-300	-130			-120	-40		

LINE CONDITION A-0001		
	°F	psig
DESIGN	160	50
OPERATING	94	0.7
MIN DESIGN	-20	50
UPSET	-	-

LINE NUMBER:- 00-A-0001-12"-1CS3